

Recommender Systems

Introduction

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Language

- course materials: English
- lectures, your presentations: probably English

Very Brief Overview

- project-based course
 - projects typically in teams (2-4 students)
 - 6 lectures (February, March)
-
- attendance registered (although not strictly compulsory)

Your Experience?

- machine learning, data mining
- information retrieval
- web implementation (PHP/Python, databases, JavaScript, ...)

A: good, B: reasonable, C: basic or none

Today

- motivation
- main notions
- course organization
- project discussion – mapping of preferences, brainstorming

Motivation

- information overload
 - many choices available
 - “the paradox of choice” (jam experiment, choice overload)
- recommender system
 - provide aid
 - set of items + user “context” $\Rightarrow$ selection of items (predicted to be “good” for the user)

Motivation

- ① What recommender systems do you know?
- ② What recommender systems would you like to have?

Examples of Applications

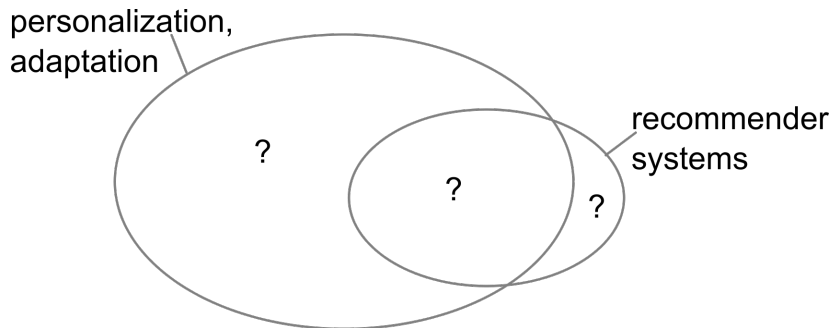
- movies, online videos
- music
- books
- software (apps)
- products in general
- people (dating, friends)
- services (restaurants, accommodation, ...)
- research articles
- jokes

Good Recommendations

What are good recommendations?

Try to think about different criteria / aspects.

Context



Recommendations, Personalization, Adaptation

- focus of the course on recommendations
- sometimes excursion into related techniques (personalization, adaptation)
 - educational applications: mastery learning

Value of Recommendations

- Netflix: 2/3 of the movies watched
- Amazon: 35% sales
- Google news: recommendations $\Rightarrow$ 38% more clickthrough

approximate, old data; up-to-date inside data are hard to get...

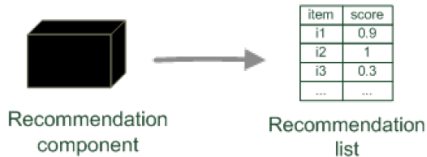
Approaches to Recommendations

Consider the previously discussed examples:

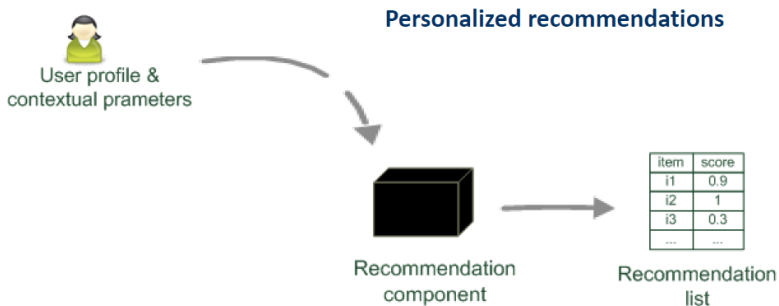
- How do the recommendations work?
- What data are used for recommendations?

Types of Recommender Systems

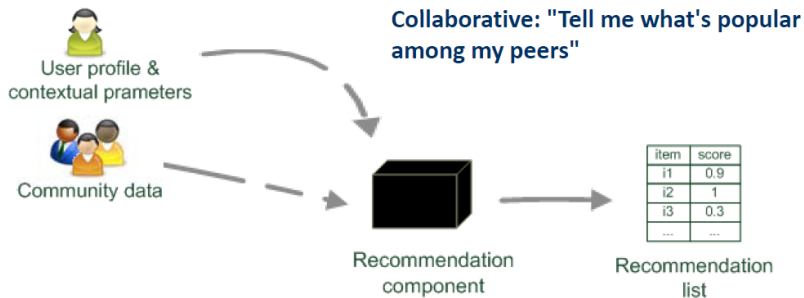
Recommender systems reduce information overload by estimating relevance



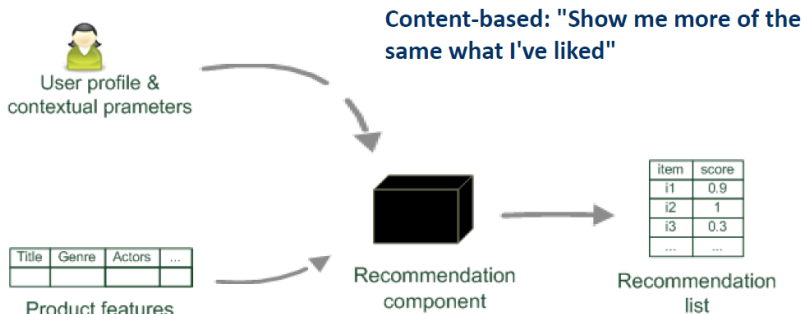
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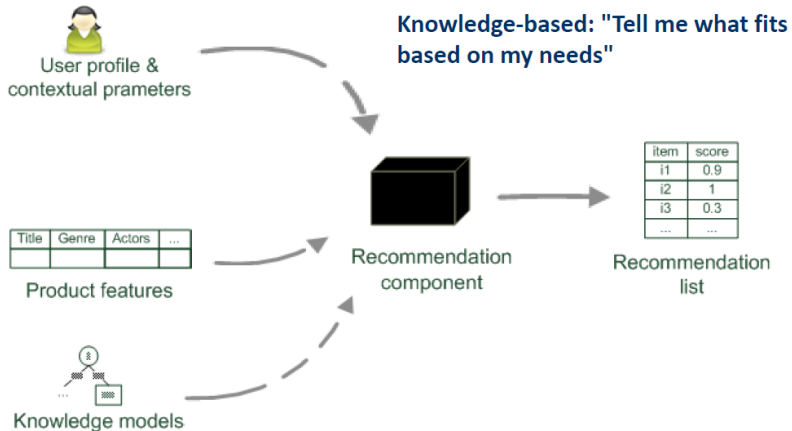
Types of Recommender Systems



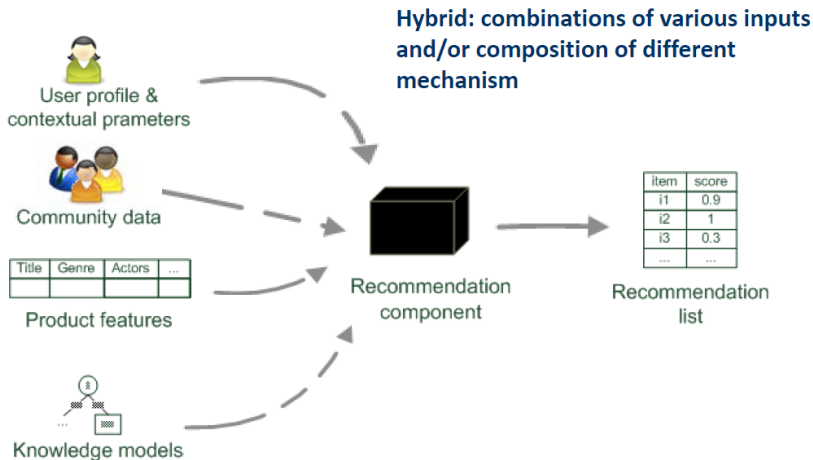
Types of Recommender Systems



Types of Recommender Systems



Types of Recommender Systems



Recommender System Functions

- provider's point of view
- user's point of view

Recommender System Functions

Provider's point of view:

- sell more items
- sell more diverse items (long tail)
- increase user satisfaction, fidelity
- better understand what users want

Long tail:



Recommender System Functions

User's point of view:

- looking for something:
 - find some good items
 - find all good items (closer to information retrieval)
 - recommend a sequence, a bundle
- just browsing
- side-effects (collaborative filtering systems):
 - express self
 - help others
 - influence others

RecSys and Information Retrieval

Information retrieval is the activity of obtaining information resources relevant to an information need from a collection of information resources. (Wikipedia)

The goal of a **Recommender System** is to generate meaningful recommendations to a collection of users for items or products that might interest them. (Melville, Sindhvani)

- RecSys and IR closely connected (many similar or analogical techniques)
- different goals:
 - IR – “I know what I’m looking for”
 - RecSys – “I’m not sure what I’m looking for”

Serendipity

- unsought finding
- unexpected, but useful result
- do not recommend items the user already knows or would find anyway, try something more interesting
- example – books:
 - I like books by Remarque, Potok, Skácel
 - recommending another book by Remarque not very useful
 - recommending Munro = serendipity

A Brief History

- 1990s – first systems (e.g., GroupLens), basic algorithms
- 1995-2000 – rapid commercialization, challenges of scale
- 2000-2005 – research explosion, mainstream applications
- 2006 – Netflix prize
- 2007 – the first Recommender Systems conference
- 2010s – applications common
- now – very active research, many applications

Netflix Prize

- Netflix – originally a video rental company
- contest: 10% improvement of the quality of recommendations
- data: user ID, movie ID, time, rating
- collaborative filtering
- prize: 1 million dollars

Recommender Systems Conference Today

- very large conference
- insight into both current research and applications

commercial sponsors RecSys conference:

DIAMOND SUPPORTER



PLATINUM SUPPORTERS



GOLD SUPPORTERS



Warning: Implementing Personalized Systems is Difficult

- (sometimes) complex algorithms
- (always) difficult debugging, testing, evaluation
 - personalization $\Rightarrow$ different behaviour for each user
 - hard to distinguish bugs and surprising results

Usefulness of Recommendations

Implementing recommendations is non-trivial.

Is it worthwhile? It depends...

- Is there “large” number of items?
- Do users know exactly what are they looking for?

Collaborative Filtering

“tell me what’s popular among my peers (=similar user)”

- one of the most often and successfully used techniques
- widely applicable, does not need any domain knowledge
- interesting analogies, metaphors, questions
 - ants, social insect: communication via pheromone
 - recommender systems: people $\sim$ ants, ratings (clicks) $\sim$ pheromone
 - between human intelligence and (good old-fashioned) artificial intelligence

Ratings

- recommender systems (particularly collaborative filtering) rely on user “ratings”
- rating of item $\sim$ how much the user likes the item
- many different forms of ratings
- what kinds of ratings do you know (can you imagine)?
- what are their advantages and disadvantages?

Ratings

- explicit
 - Likert scale (5 stars), like/dislike
 - require additional effort from users
- implicit
 - click through rate, buying an item, visiting a page, viewing a video, dwell time
 - easier to collect, less precise
 - more “honest” (Netflix example: highly rated vs watched)

Recommended reading: <https://www.wired.com/2013/08/qq-netflix-algorithm/>

“We know that many of the ratings are aspirational rather than reflecting your daily activity.”

Potential Downside

- serving “low instincts” instead of “high aspirations” ?
- news, optimizing clicks:
 - sex, tragedy, fear, celebrity
 - thorough analysis, complex problems

Potential Downside II

personalization in general, collaborative filtering specifically

- “filter bubbles”
- news, social media
- users only see what they are expected to like
 - good for business (in the short term)
 - potentially bad (in the long term) for users and society

Downsides: What does it mean for us?

- do not “throw away” collaborative filtering techniques
- be aware of the limitations
- try to address limitations in suitable way (depending on the application)

Goals, Evaluation

- What is the goal of the system?
- How do we evaluate a recommender system?
- What is a “good” recommender system?
- How do we quantify the performance?

important topics of the course

RecSys and Educational Domain

- learning materials – direct application
- learning task, exercises:
 - users $\sim$ students
 - items $\sim$ learning tasks
 - ratings $\sim$ performance (correctness of answers, problem solving times)

Personalization in Education

- adaptive learning, personalized learning, ...
- well-known:
 - open systems: Khan Academy, Duolingo
 - commercial companies: Pearson, Knewton
- local, my experience:
 - Adaptive Learning group:
www.fi.muni.cz/adaptivelearning/
 - Umime (umimeto.org)
 - research spin-off, product used by over 1000 schools
 - practically used recommendation algorithm

Course Organization

- February, March (6 weeks)
 - lectures: main notions of the field
 - discussions: relations of notions to your projects
- April
 - work on projects
 - individual consultations
- May
 - presentation of projects

Focus of This Course

- practical experience
- collaborative filtering
- educational applications
- evaluation (illustration of methodological issues relevant not just for RecSys)

focus on consultations / discussions (good lectures/materials available online)

Preliminary Schedule – Lectures

- February 20: Collaborative filtering
- February 27: Other recommendation techniques
- March 5: Evaluation
- March 12: Educational recommender systems, practical experiences
- March 19: Practical aspects; Case studies

Prerequisites

- programming
- math (basic linear algebra, statistics)
- basics of machine learning (not strictly necessary)

(depends also on the choice of project)

Materials, Sources

- Introduction to Recommender Systems book
 - <http://www.recommenderbook.net/>
 - slides freely available – more details than in course slides
- Recommender Systems Handbook
 - electronic version available from MU
- Video lectures: Coursera, Machine learning summer school

(links at the course web page)

Projects

2 basic options:

- “application”: development of a simple recommender system
- “research”: implementation and experimental evaluation of algorithms used by recommender systems

many different “hybrids” possible (e.g., extension / analysis of data from your own system)

“Application”: System Development

- team project (1-4 students)
- goal: build a simple recommender system
- realization
 - simple web page implementation (e.g., Python / MySQL / JavaScript)
 - console application

*note: consultations will be about “recommendation topics”,
not about web page implementation*

Ideas for Simple Recommender System

- “short text” recommendations: jokes, quotes, poetry, baby names, recipes
- travel, “local” recommendations (Brno): restaurants, cultural events, places, holiday locations, tourist attractions, geocaching
- educational recommendations: courses (MU, MOOC), foreign language vocabulary, learning materials
- product recommendation (specialized for a particular domain): board games, beers, specific movie genre

Typical Steps

- clarification of the purpose (for whom? why?), specific aspects of the domain, hypothetical business model
- getting/generating data
- basic analysis of data
- implementation of a simple web system
- design and implementation of several recommendation techniques
- evaluation
- presentation

Focus of Project

- “simple domains” (e.g., jokes, English vocabulary)
 - several recommendation algorithms (different types)
 - collection of your own data (ratings, feedback), analysis, evaluation
- “complex domains” (e.g., extension of an existing system)
 - analysis of existing data (what can we use for recommendations)
 - “design” of recommendations, formulation of aims, ...
 - evaluation: proposal, first steps

Advice I

- prefer larger team (3 or 4 students)
- clear division of tasks, responsibilities
- use version control system (GitHub, gitlab.fi.muni.cz, ...)

Advice II

experience from previous years:

- prefer something rather simple, but done well, focus on recommendation aspects
- ambitious projects often:
 - too much time on technical aspects (getting and cleaning data, implementation infrastructure)
 - little time left for recommendations

“Research” : Models, Evaluation

individual project or group in (mainly) “competitive mode”

- use existing data with ratings (movies, books, ...)
- develop a model for predicting user ratings
- evaluate the model, visualize results

requirements: data analysis (Python recommended),
implementation of machine learning techniques

Projects from Previous Years

- products: board games, video games, wine, beer, PC parts
- funny quotes, jokes, recipes, blog posts, jobs, anime/manga, geocaching, linux applications
- educational resources, English vocabulary, MU courses, master theses
- analysis of data from existing systems: movies, music, board games, blog system, geography learning
- implementation of techniques into a real e-shop

Course Deliverables

- source code with basic documentation
- presentation
- individual report (2-3 pages)
 - description of individual contribution to the project
 - connection with course topics
 - discussion of related research papers

Colloquium – Requirements

standard way:

- active participation during semester
- interesting project, presentation, report

special cases (poor attendance, weak project, unclear contribution to the project, etc):

- revision of the project
- individual “examination” (discussion) at the end of semester

Discussion

- questions
- your project ideas
- potential groups